

Isogeny graphs of ordinary elliptic curves over finite fields

Max Kortge, supervised by Alex Ghitza

The University of Melbourne

Introduction

The set of isomorphism classes of elliptic curves can be viewed as the vertices of a graph, called the isogeny graph, where an edge between curves exists if and only if there exists an isogeny between them.

There has been great interest in understanding the structure of isogeny graphs over finite fields. The isogeny graph can be split into two disjoint components based on whether the elliptic curves are **ordinary** or **supersingular**. The supersingular graph has a complicated structure and has thus been studied extensively for its cryptographic applications, whereas the ordinary graph has a fairly regular structure. A further development has been to add extra structure to the elliptic curves, called a level N structure, and study the isogeny graphs that arise.

In this poster we will explore the structure of ordinary isogeny graphs, both with and without level structure, over finite fields.

Complex beginnings

The **Uniformisation Theorem** gives an explicit correspondence between elliptic curves over $\mathbb{C}$ and complex tori $\mathbb{C}/L$ defined by lattices $L \subseteq \mathbb{C}$. In particular, homothety of lattices corresponds with isomorphisms of elliptic curves.

For any lattice $L = [\omega_1, \omega_2]$ in $\mathbb{C}$, L is homothetic to $[1, \tau]$ where $\tau = \pm\omega_2/\omega_1$ is chosen so that τ lies in the upper half-plane $\mathcal{H} := \{z \in \mathbb{C} \mid \Im(z) > 0\}$. We thus restrict our attention to $\tau \in \mathcal{H}$.

The **Eisenstein series of weight $2k$** , where $k \geq 2$ is an integer, is

$$G_{2k}(\tau) = \sum_{(m,n) \in \mathbb{Z}^2 \setminus \{(0,0)\}} \frac{1}{(m + n\tau)^{2k}}.$$

Using this, we define the **j -function** $j : \mathcal{H} \rightarrow \mathbb{C}$ by

$$j(\tau) = 1728 \frac{g_2(\tau)^3}{g_2(\tau)^3 - 27g_3(\tau)^2},$$

where $g_2(\tau) := 60G_4(\tau)$ and $g_3(\tau) := 140G_6(\tau)$. It is a holomorphic function that is 1-periodic and invariant under the map $\tau \mapsto -1/\tau$.

Further, the j -function coincides with the j -invariant of an elliptic curve. Since (over an algebraically closed field) the j -invariant characterises elliptic curves up to isomorphism, if we want to understand elliptic curves over $\mathbb{C}$ we should study the j -function.

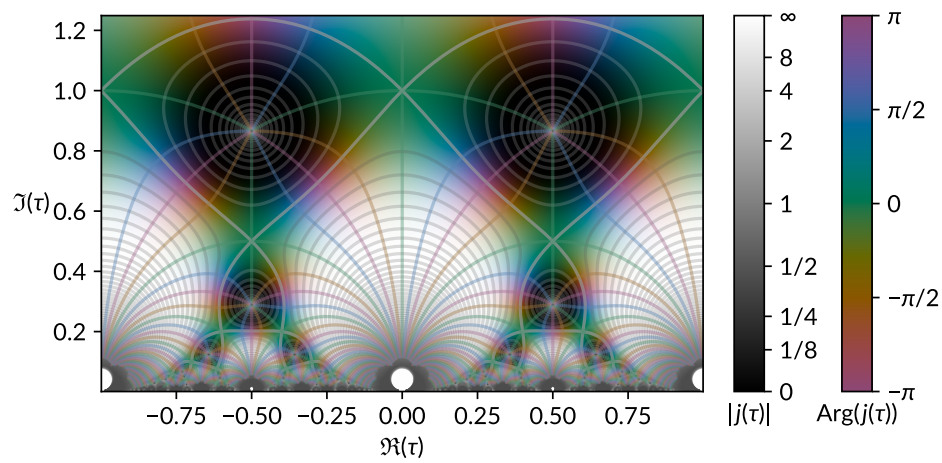


Figure 1: Domain colouring plot of the j -function. The emphasised contour is $|j(\tau)| = 1$.

A modular detour

The **modular group** is

$$\Gamma := \text{SL}_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{Z}, ad - bc = 1 \right\}.$$

It acts on the left on $\mathcal{H}$ by linear fractional transformations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau = \frac{a\tau + b}{c\tau + d}.$$

The j -function is invariant under this action: for $\tau, \tau' \in \mathcal{H}$, $j(\tau) = j(\tau')$ if and only if $\tau' = \gamma \cdot \tau$ for some $\gamma \in \Gamma$. Thus we can narrow our study of the j -function to the Γ -equivalence classes of $\mathcal{H}$ (the orbits of $\mathcal{H}$ under the action of Γ). We denote the quotient $\Gamma \backslash \mathcal{H}$ by $Y(1)$ and call it a **modular curve**.

We should now pick a set of representatives for each equivalence class, that is, a fundamental domain for Γ . We use the set

$$\mathcal{F} := \left\{ \tau \in \mathcal{H} \mid |\Re(\tau)| \leq \frac{1}{2} \text{ and } |\tau| \geq 1 \right\},$$

with sides identified as shown in Figure 2.

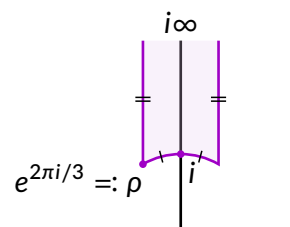


Figure 2: $\mathcal{F}$.

Notice however that $\mathcal{F}$ is not compact. To remedy this, we compactify it by adding a point at infinity, $\infty := i\infty$. We still want Γ to act on this space and as

$$\lim_{\Im(\tau) \rightarrow \infty} \frac{a\tau + b}{c\tau + d} = \frac{a}{c} \in \mathbb{Q},$$

we need to add $\mathbb{Q}$. Hence we define the extended upper half-plane as the set $\mathcal{H}^* = \mathcal{H} \cup \mathbb{Q} \cup \{\infty\} = \mathcal{H} \cup \mathbb{P}^1(\mathbb{Q})$ and the modular curve $X(1) := \Gamma \backslash \mathcal{H}^*$, which can be identified with the fundamental domain $\mathcal{F}^* = \mathcal{F} \cup \{\infty\}$. $X(1)$ is a connected Hausdorff space and can be endowed with a complex structure, making it a Riemann surface.

In defining the complex structure, we must take into account the stabilisers of $\tau \in \mathcal{F}^*$ in Γ . Most points in $\mathcal{F}^*$ are fixed only by $\pm I$, but there are three points where this is not the case:

- i , which corresponds to the j -invariant 1728, has $\text{Stab}(i) \cong \mathbb{Z}/4\mathbb{Z}$;
- ρ , which corresponds to the j -invariant 0, has $\text{Stab}(\rho) \cong \mathbb{Z}/6\mathbb{Z}$; and
- ∞ has $\text{Stab}(\infty) \cong \mathbb{Z}$.

For $\tau \in \mathcal{F}$, the automorphism group of the elliptic curve with j -invariant $j(\tau)$ is isomorphic to $\text{Stab}(\tau)$. As we shall see later, this is what makes the j -invariants 0 and 1728 special.

More modular curves

For $N \in \mathbb{Z}^+$, the **principal congruence subgroup of level N** is

$$\Gamma(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}) \mid \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}.$$

A **congruence subgroup** (of level N) is a subgroup of $\text{SL}_2(\mathbb{Z})$ that contains $\Gamma(N)$. Two of particular interest are

$$\Gamma_1(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}) \mid \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}; \text{ and}$$

$$\Gamma_0(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}) \mid \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{N} \right\}.$$

As we did above for $Y(1)$ and $X(1)$, one can similarly define the modular curves $Y(N)$, $Y_1(N)$, $Y_0(N)$ for $\Gamma(N)$, $\Gamma_1(N)$, $\Gamma_0(N)$ respectively and their compactifications $X(N)$, $X_1(N)$, $X_0(N)$, all of which are Riemann surfaces.

The modular equation

Consider the minimal polynomial $\Phi_N(Y)$ of $j(N\tau)$ over the function field $\mathbb{C}(j)$. Replacing the occurrences of j in $\Phi_N(Y)$ with a new variable X , we get the **modular polynomial $\Phi_N(X, Y)$** . It has integer coefficients and, in both variables, is symmetric and has degree $N + 1$.

We say that an isogeny is **cyclic** if its kernel is a cyclic group. For $\tau \in \mathcal{H}$ there exists a cyclic isogeny, whose kernel has order N , between the elliptic curves with j -invariants $j(\tau)$ and $j(N\tau)$. The functions $j(\tau)$ and $j(N\tau)$ define a bijection $\tau \mapsto (j(\tau), j(N\tau))$ from $Y_0(N)$ to the affine curve $\Phi_N(X, Y) = 0$. We call $\Phi_N(X, Y) = 0$ the **modular equation** and say that $X_0(N)$ is a **moduli space** for the **moduli problem** of determining equivalence classes of pairs (E, C) , where E is an elliptic curve and $C \subseteq E$ is a cyclic subgroup of order N .

Any isogeny can be decomposed into a composition of prime degree isogenies, and every N -isogeny is cyclic when N is prime. Thus two elliptic curves E and E' are such that $\Phi_N(j(E), j(E')) = 0$ if and only if they are related by an N -isogeny! This interpretation holds over any field, in particular over finite fields (thanks in part to $\Phi_N(X, Y) \in \mathbb{Z}[X, Y]$).

Isogeny volcanoes

With the modular equation in hand, we can now define for prime ℓ the graph of ℓ -isogenies over a field k with $\text{char } k \neq \ell$. The **ℓ -isogeny graph $\mathcal{X}_\ell(k)$** is the directed graph with vertex set k and m edges (j, j') present whenever j' is a root of $\Phi_\ell(j, Y)$ with multiplicity m . Due to the existence of an isogeny's dual, (j, j') is an edge in $\mathcal{X}_\ell(k)$ if and only if (j', j) is also an edge. When $j, j' \neq 0, 1728$, these edges have the same multiplicity, and so $\mathcal{X}_\ell(k)$ can be considered (for the most part) an undirected graph.

We define $\mathcal{G}_\ell(k)$ to be the subgraph of $\mathcal{X}_\ell(k)$ containing only the vertices of ordinary elliptic curves. Over a finite field $k = \mathbb{F}_q$, the structure of $\mathcal{G}_\ell(\mathbb{F}_q)$ can be completely determined. Each connected component of $\mathcal{G}_\ell(\mathbb{F}_q)$, not containing the vertices 0 and 1728, looks like a volcano!

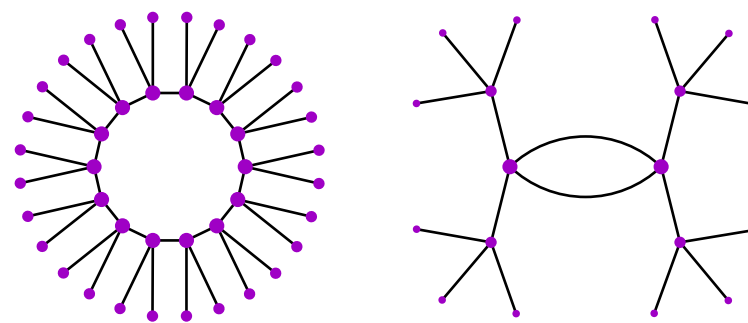


Figure 3: Two connected components (volcanoes) of $\mathcal{G}_3(\mathbb{F}_{1009})$.

More precisely, they are an **ℓ -volcano V** , a connected undirected graph whose vertices admit a partition into one or more **levels** $V_0, \dots, V_d$ such that

- the subgraph on V_0 (the **crater**) is a regular graph of degree at most 2;
- for $0 < i < d$, each vertex in V_i has degree $\ell + 1$; and
- for $0 < i \leq d$, each vertex in V_i has exactly one neighbour in level V_{i-1} .

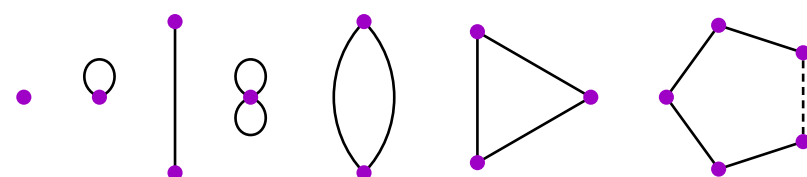


Figure 4: All possible craters.

Pathological volcanoes

Above we excluded the components containing vertices 0 and 1728 in our definition of a volcano. This has to do with how an edge (j, j') , for one of j or j' equal to 0 or 1728, does not necessarily have the same multiplicity as its reverse (j', j) .

Consider an ℓ -isogeny $\varphi : E \rightarrow E'$. Each $a \in \text{Aut}(E)$ gives another ℓ -isogeny $\varphi \circ a$ (which may or may not be distinct from φ i.e., have equal kernels), and if $\text{Aut}(E) = \text{Aut}(E')$ there exists $a' \in \text{Aut}(E')$ such that $\varphi \circ a = \hat{\varphi} \circ a'$. But if $\text{Aut}(E)$ is strictly larger than $\text{Aut}(E')$, there cannot exist distinct $a' \in \text{Aut}(E)$ such that $\varphi \circ a = \hat{\varphi} \circ a'$ for every $a \in \text{Aut}(E)$. Hence there could be a discrepancy in the number of isogenies in each direction.

Recall that the automorphism group of an elliptic curve is $\{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z}$, except for curves with j -invariant 1728 or 0, which have automorphism groups $\{\pm 1, \pm i\} \cong \mathbb{Z}/4\mathbb{Z}$ and $\{\pm 1, \pm \rho, \pm \rho^2\} \cong \mathbb{Z}/6\mathbb{Z}$ respectively. The automorphism -1 fixes every cyclic subgroup of an elliptic curve, and so the isogeny $\varphi \circ [-1]$ is considered the same as φ , since they have the same kernel. However, for j -invariant 0 (1728) the isogenies φ , $\varphi \circ \rho$, $\varphi \circ \rho^2$ (φ , $\varphi \circ i$) all have different kernels. So an edge from an elliptic curve with j -invariant 0 (1728) will have multiplicity 3 (2), while its reverse edge will have multiplicity 1!

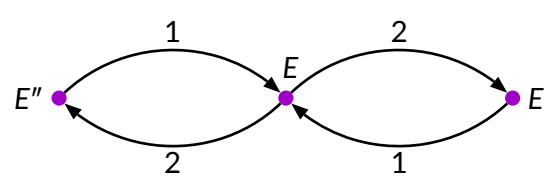


Figure 5: Connected component of $\mathcal{G}_3(\mathbb{F}_{1009})$ containing the elliptic curve E with j -invariant 1728. The numbers next to the edges are their multiplicities.

Level structure

Previously we mentioned how $X_0(N)$ is a moduli space for the problem of determining pairs (E, C) , where E is an elliptic curve and C is a cyclic subgroup of order N . What about the other congruence subgroups we defined earlier? Their modular curves are also moduli spaces for a moduli problem.

First we define for an elliptic curve E its **m -torsion subgroup $E[m] := \{P \in E \mid mP = O\}$** . Each moduli problem, given below, is related to the m -torsion subgroup.

$$\begin{aligned} X(N) &\longleftrightarrow \left\{ \begin{array}{l} \text{isomorphism classes of triples } (E, P, Q) \\ \text{where } \langle P, Q \rangle = E[N] \end{array} \right\} \\ X_1(N) &\longleftrightarrow \left\{ \begin{array}{l} \text{isomorphism classes of pairs } (E, P) \\ \text{where } P \in E[N] \text{ has order } N \end{array} \right\} \\ X_0(N) &\longleftrightarrow \left\{ \begin{array}{l} \text{isomorphism classes of pairs } (E, \langle P \rangle) \\ \text{where } \langle P \rangle \subseteq E[N] \text{ is cyclic of order } N \end{array} \right\} \end{aligned}$$

Two tuples (E, γ) and (E', γ') are equivalent if there is an isomorphism $E \cong E'$ that takes γ to γ' .

When $N = 1$, $X(1) = X_1(1) = X_0(1)$ and these are simply moduli spaces for the problem of determining elliptic curves up to isomorphism.

Adding level structure

Now we shall use the above to consider the isogeny graph formed when we add additional structure to an elliptic curve. The structure to be imposed is called a **level structure** and is defined as follows for an elliptic curve E

- a $\Gamma(N)$ -level structure (or full level structure) is an ordered pair of points $P, Q \in E[N]$ which are a basis for $E[N]$;
- a $\Gamma_1(N)$ -level structure is a point $P \in E[N]$ of order N ; and
- a $\Gamma_0(N)$ -level structure is a cyclic subgroup $\langle P \rangle \subseteq E[N]$ of order N .

We denote the ℓ -isogeny graphs obtained by adding a $\Gamma(N)$ -, $\Gamma_1(N)$ -, $\Gamma_0(N)$ -level structure to each vertex in $\mathcal{X}_\ell(k)$, up to the equivalence relation defined earlier, by $\mathcal{X}_\ell^{\Gamma(N)}(k)$, $\mathcal{X}_\ell^{\Gamma_1(N)}(k)$, $\mathcal{X}_\ell^{\Gamma_0(N)}(k)$ respectively. We also denote $\mathcal{G}_\ell^*(k)$ to be the subgraph of $\mathcal{X}_\ell^*(k)$ containing only ordinary elliptic curves.

An edge from (E, γ) to (E', γ') exists when there exists an ℓ -isogeny $\varphi : E \rightarrow E'$ such that $\varphi(\gamma) = \gamma'$. However, unlike in $\mathcal{X}_\ell(k)$, an isogeny's dual will not necessarily give rise to an edge in the opposite direction. Indeed, for P a $\Gamma_1(N)$ -level structure and an ℓ -isogeny φ , $(\hat{\varphi} \circ \varphi)(P) = \ell P \neq P$ in general. Thus we consider $\mathcal{X}_\ell^*(k)$ to be a directed graph.

Why add more structure?

Recall how, when adding a complex structure to $Y(1)$, we had to take into account three points whose stabilisers were not isomorphic to $\mathbb{Z}/2\mathbb{Z}$. This meant the elliptic curves with j -invariants 0 and 1728 had extra automorphisms. By adding level structure to a space we "rigidify" its geometry, reducing its symmetries. Concretely, this means for a given point the set of elements in the congruence subgroup which fix any given point will change. This will correspond with a change in the automorphism groups of the isomorphism classes of our elliptic curves with level structure.

Consider a $\Gamma(N)$ -level structure. When $N = 2$, every point in $\mathcal{H}$ has stabiliser isomorphic to $\mathbb{Z}/2\mathbb{Z}$. Hence their corresponding elliptic curves will have automorphism group isomorphic to $\mathbb{Z}/2\mathbb{Z}$. Like in the isogeny graph without level structure, the automorphism -1 does not give any additional distinct isogenies. Thus the problems that arise with j -invariants 0 and 1728 have vanished! Figure 6 gives an example of what this means for j -invariant 1728. Notice that the graph is no longer an ℓ -volcano.

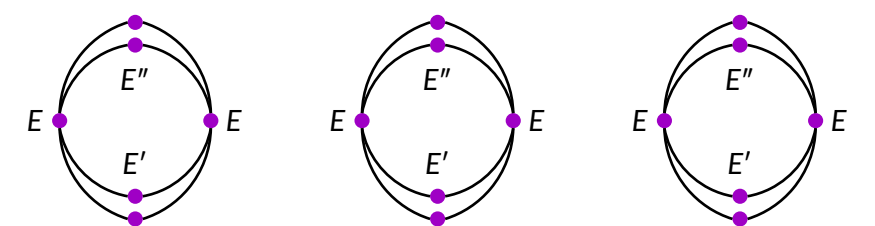


Figure 6: Components of $\mathcal{G}_3^{\Gamma(2)}(\mathbb{F}_{1009})$ that contain the elliptic curve E with j -invariant 1728 (cf. Figure 5).

Further, when $N \geq 3$, every point in $\mathcal{H}$ has trivial stabiliser, and thus their corresponding elliptic curves have trivial automorphism groups.

In future work, we intend to try to further our understanding of how the structure of the isogeny graph changes when we add level structure to it.

Acknowledgements

I would like to thank my supervisor Alex Ghitza for his support throughout this project, and Dhruv Gupta whom I worked together with on this project. I would also like to thank the School of Mathematics and Statistics for the opportunity to participate in this program.

References

- [1] Sutherland A. Isogeny volcanoes. The Open Book Series 2013;1:507–30. <https://doi.org/10.2140/obs.2013.1.507>.
- [2] Lei A, Müller K. On ordinary isogeny graphs with level structures. Expositiones Mathematicae 2024;42:125589. <https://doi.org/10.1016/j.exmath.2024.125589>.
- [3] Perrin D, Voloch JF. Ordinary Isogeny Graphs with Level Structure 2024. <https://doi.org/10.48550/arXiv.2411.02732>.
- [4] Bambury H, Campagna F, Pazuki F. Ordinary isogeny graphs over $\mathbb{F}_p$: the inverse volcano problem 2022. <https://doi.org/10.48550/arXiv.2210.01086>.
- [5] Fouquet M, Morain F. Isogeny volcanoes and the SEA algorithm. Algorithmic Number Theory (Sydney, 2002) 2002;2369:276–91. https://doi.org/10.1007/3-540-45455-1_23.
- [6] Kohel DR. Endomorphism rings of elliptic curves over finite fields. ProQuest LLC, Ann Arbor, MI; 1996.